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PREMIUM NOTES

UPSC — Geography Notes

UPSC Test Series — StuDy

High-yield facts, formulas and trap-radar — made for the real exam hall.

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UPSC — Geography Notes

Physical Geography — Processes

- Plate tectonics: Himalaya = continent–continent convergence (Indian plate moving NE ~5 cm/yr); earthquakes concentrated along plate boundaries; Wadati–Benioff zone = subduction seismic belt.
- Rocks: Igneous (primary — basalt Deccan traps from fissure eruptions ~66 mya), Sedimentary (stratified, fossils), Metamorphic (marble from limestone, gneiss from granite, schist). Rock cycle links all three.
- Landforms: river stages — youthful (V-valley, rapids, waterfalls), mature (meanders), old (oxbow lakes, deltas). Karst (limestone — caves, sinkholes — Meghalaya caves). Aeolian: barchans (crescent dunes), loess. Glacial: moraines, drumlins, U-valleys, fjords.
- Atmosphere: layers — Troposphere (weather, lapse rate 6.5°C/km), Stratosphere (ozone), Mesosphere (meteors burn), Thermosphere/Ionosphere (radio reflection), Exosphere. Jet streams: sub-tropical westerly (drives western disturbances), tropical easterly (monsoon link).
- Ocean: salinity highest in Dead Sea/lake Van; currents — warm (Kuroshio, Gulf Stream) vs cold (Oyashio, Labrador, Benguela, Humboldt/Peru — Atacama dryness); El Niño = warm water off Peru coast; tides = moon > sun gravity (spring/neap).

Indian Geography — Systems

- Drainage patterns: dendritic (plains), trellis (folded — old Himalaya streams), radial (Amarkantak), antecedent rivers (Indus, Sutlej, Brahmaputra — older than Himalaya, cut gorges).
- Monsoon mechanics: ITCZ shift, Tibetan plateau heating (thermal low), Somali jet, Mascarene high, ENSO & IOD (Indian Ocean Dipole — positive IOD can offset weak El Niño years). Break monsoon = rain shifts to Himalayan foothills/Brahmaputra.
- Soil orders (ICAR 8): Alluvial, Black, Red, Laterite, Arid, Forest/Mountain, Peaty/Marshy, Saline/Alkaline. Soil health: NPK ideal ratio 4:2:1 (India skewed towards N — urea subsidy effect).
- Natural vegetation: Tropical evergreen (>200 cm rain — Western Ghats, NE — ebony, mahogany, rosewood), moist deciduous (100–200 cm — teak, sal), dry deciduous (70–100 cm — tendu), thorn (<50 cm — acacia), montane, mangrove (Sundarbans largest — Sundari tree; Bhitarkanika second).
- Resources & industry location: coal (Jharia richest, Raniganj oldest), iron ore (Odisha–Chhattisgarh–Karnataka belt; Bailadila high grade), bauxite (Odisha top), mica (Koderma). Steel plants & their logic: Bhilai (Russian collab), Rourkela (German), Durgapur (British), Bokaro (Russian) — near coal/iron + water + rail.

Environment & Ecology

- Biodiversity: India = 4 of 36 global hotspots (Himalaya, Indo-Burma, Western Ghats–Sri Lanka, Sundaland). Levels: genetic, species, ecosystem. Alpha (within habitat), Beta (between), Gamma (regional) diversity.
- Protected areas hierarchy: National Park (107) > Sanctuary (573) > Conservation/Community Reserves; Biosphere Reserves (18 — Nilgiri first 1986; 12 in UNESCO MAB); Tiger Reserves (Project Tiger 1973 — NTCA statutory under WLPA amendment 2006; Corbett first NP 1936).
- Conventions: Ramsar (1971, wetlands — India 90+ sites, TN highest), CITES (trade in species), CBD (1992 — Nagoya Protocol ABS, Aichi/Kunming-Montreal targets 30×30),

UNFCCC (1992) → Kyoto (1997, binding developed) → Paris (2015, NDCs; India: net zero 2070, 50% non-fossil capacity by 2030), Stockholm (POPs), Basel (hazardous waste), Minamata (mercury).

- Species files: Bengal tiger (apex, umbrella), Asiatic lion (Gir only — population rising, Barda proposed second home), Great Indian Bustard (Critically Endangered — Rajasthan state bird), Gangetic dolphin (National Aquatic Animal), Olive Ridley (Gahirmatha mass nesting — arribada), Red Panda (Eastern Himalaya), Hangul (Kashmir stag — Dachigam).
- Pollution & health: AQI categories (0–500; PM2.5/PM10, NO_2, SO_2, CO, O_3, NH_3, Pb); stubble burning (PUSA decomposer), Bharat Stage VI (2020), NCAP (2019). Waste rules: e-waste, plastic (single-use ban 2022), solid waste 2016.

Mapping — PYQ Locations

- Straits: Malacca (Malaysia–Indonesia), Hormuz (oil chokepoint — Iran/Oman), Bab-el-Mandeb (Red Sea–Gulf of Aden), Bering (Russia–Alaska), Gibraltar, Taiwan, Mozambique Channel.
- Conflict/news regions: Gaza Strip, West Bank, Golan Heights, Donbas, South China Sea (nine-dash line), Senkaku/Diaoyu, Kuril Islands, Nagorno-Karabakh.
- Indian mapping: Siliguri Corridor ('Chicken's Neck'), Doklam (Bhutan–China–India trijunction), Galwan (Ladakh, LAC), Sir Creek (Gujarat–Sindh dispute), Kalapani–Lipulek (Nepal), Katchatheevu (Sri Lanka).

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